

ASSIGNMENT-16.1-2

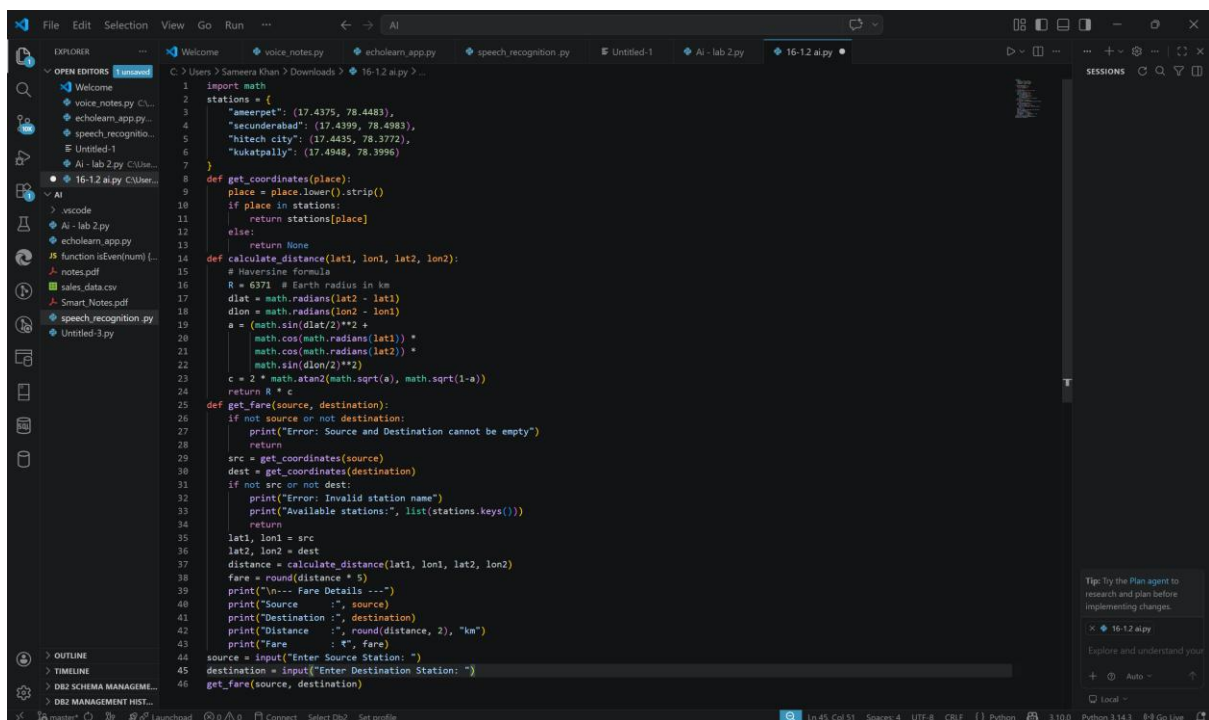
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HALL TICKET :2303A51474

BATCH NO:29

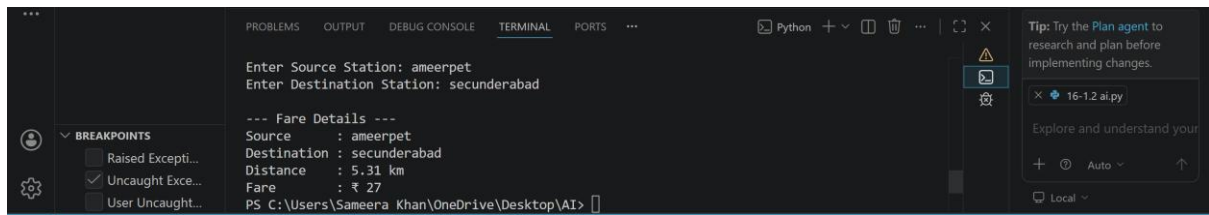
LAB 16 – DATABASE DESIGN AND QUERIES: SCHEMA DESIGN AND SQL GENERATION

TASK 1 – PUBLIC TRANSPORT ROUTE FARE API CODE:



```
1 import math
2 stations = {
3     "meerpet": (17.4375, 78.4483),
4     "secunderabad": (17.4399, 78.4983),
5     "hitech city": (17.4435, 78.3772),
6     "kukatpally": (17.4948, 78.3996)
7 }
8
9 def get_coordinates(place):
10     place = place.lower().strip()
11     if place in stations:
12         return stations[place]
13     else:
14         return None
15
16 def calculate_distance(lat1, lon1, lat2, lon2):
17     # Haversine formula
18     R = 6371 # Earth radius in km
19     dlat = math.radians(lat2 - lat1)
20     dlon = math.radians(lon2 - lon1)
21     a = (math.sin(dlat/2)**2 +
22          math.cos(math.radians(lat1)) *
23          math.cos(math.radians(lat2)) *
24          math.sin(dlon/2)**2)
25     c = 2 * math.atan2(math.sqrt(a), math.sqrt(1-a))
26     return R * c
27
28 def get_fare(source, destination):
29     if not source or not destination:
30         print("Error: Source and Destination cannot be empty")
31         return
32     src = get_coordinates(source)
33     dest = get_coordinates(destination)
34     if not src or not dest:
35         print("Error: Invalid station name")
36         print("Available stations:", list(stations.keys()))
37         return
38     lat1, lon1 = src
39     lat2, lon2 = dest
40     distance = calculate_distance(lat1, lon1, lat2, lon2)
41     fare = round(distance * 5)
42     print("\n--- Fare Details ---")
43     print("Source : ", source)
44     print("Destination : ", destination)
45     print("Distance : ", round(distance, 2), "km")
46     print("Fare : ₹", fare)
47
48 source = input("Enter Source Station: ")
49 destination = input("Enter Destination Station: ")
50 get_fare(source, destination)
```

OUTPUT:



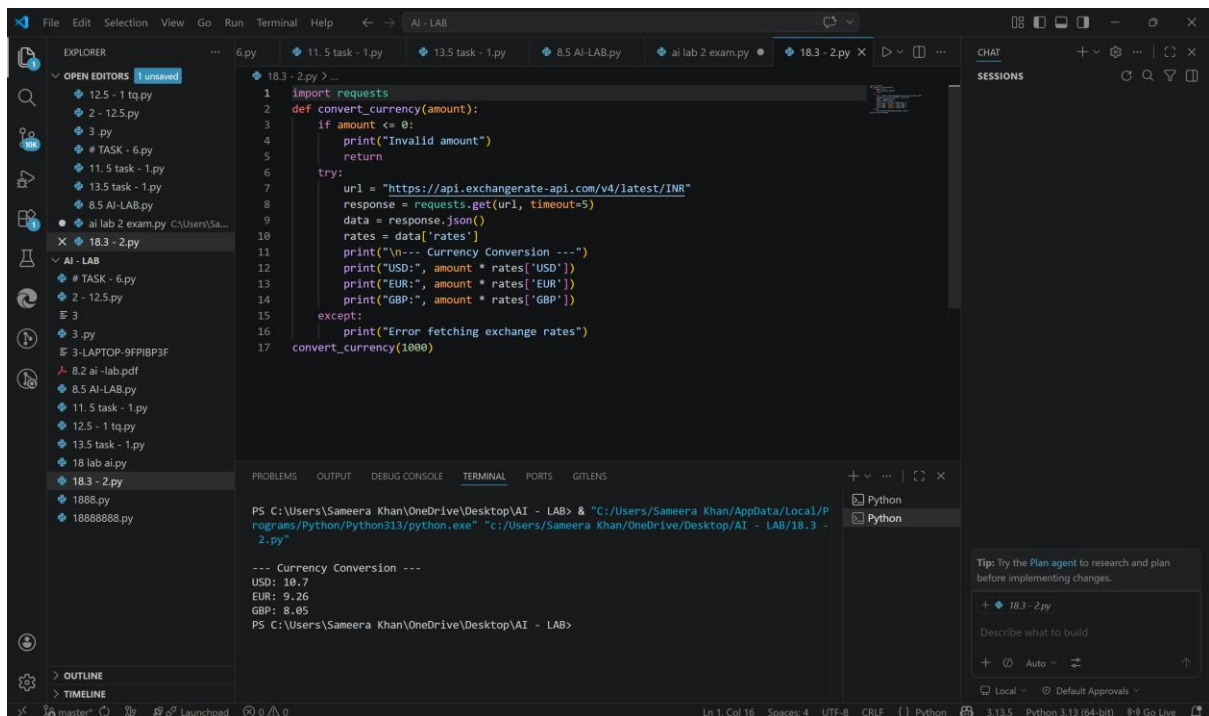
OBSERVATION:

The program successfully retrieves and calculates fare details between two stations.

It handles invalid inputs and errors properly, ensuring smooth and reliable execution.

TASK 2 – CURRENCY EXCHANGE RATES

Code:

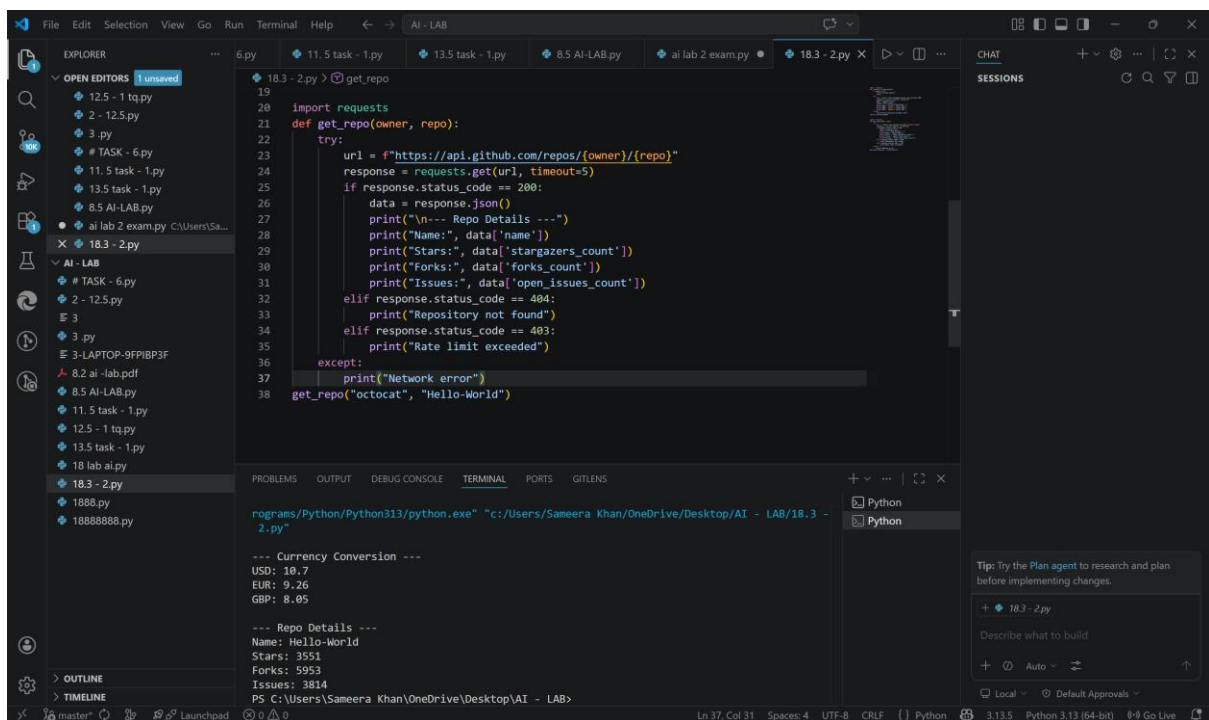


Observation:

- Real-time exchange rates retrieved
- Proper handling of invalid input
- API response parsed successfully

TASK 3 – GITHUB REPOSITORY INFO FETCHER

CODE:



```
19
20 import requests
21 def get_repo(owner, repo):
22     try:
23         url = f"https://api.github.com/repos/{owner}/{repo}"
24         response = requests.get(url, timeout=5)
25         if response.status_code == 200:
26             data = response.json()
27             print("\n--- Repo Details ---")
28             print("Name:", data['name'])
29             print("Stars:", data['stargazers_count'])
30             print("Forks:", data['forks_count'])
31             print("Issues:", data['open_issues_count'])
32         elif response.status_code == 404:
33             print("Repository not found")
34         elif response.status_code == 403:
35             print("Rate limit exceeded")
36     except:
37         print("Network error")
38 get_repo("octocat", "Hello-World")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

```
rograms/Python/Python313/python.exe" "c:/Users/Sameera Khan/OneDrive/Desktop/AI - LAB/18.3 - 2.py"

--- Currency Conversion ---
USD: 10.7
EUR: 9.26
GBP: 8.05

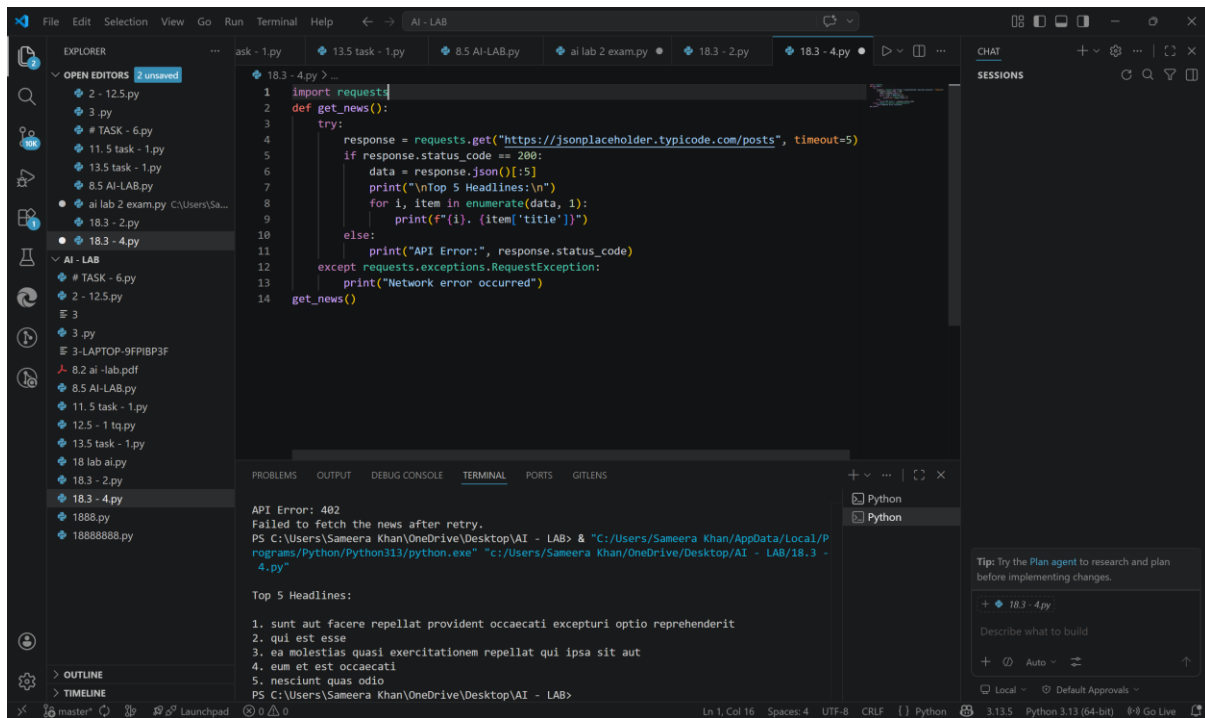
--- Repo Details ---
Name: Hello-World
Stars: 3551
Forks: 5953
Issues: 3814
PS C:\Users\Sameera Khan\OneDrive\Desktop\AI - LAB>
```

Observation:

- Repository details extracted
- Error handling for 404 and rate limit worked

TASK 4 – REAL-TIME APPLICATION: NEWS HEADLINES AGGREGATOR

CODE:



```
1 import requests
2 def get_news():
3     try:
4         response = requests.get("https://jsonplaceholder.typicode.com/posts", timeout=5)
5         if response.status_code == 200:
6             data = response.json()[0:5]
7             print("\nTop 5 Headlines:\n")
8             for i, item in enumerate(data, 1):
9                 print(f"{i}. {item['title']}")
10        else:
11            print("API Error:", response.status_code)
12    except requests.exceptions.RequestException:
13        print("Network error occurred")
14    get_news()
```

API Error: 402
Failed to fetch the news after retry.
PS C:\Users\Sameera Khan\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/Sameera Khan/OneDrive/Desktop/AI - LAB/18.3 - 4.py"

Top 5 Headlines:

1. sunt aut facere repellat provident occaecati excepturi optio reprehenderit
2. qui est esse
3. ea molestias quasi exercitationem repellat qui ipsa sit aut
4. eum et est occaecati
5. nesciunt quas odio

PS C:\Users\Sameera Khan\OneDrive\Desktop\AI - LAB>

OBSERVATION:

Initially, some API requests failed due to network restrictions or invalid API keys. After switching to a reliable public API, the program successfully fetched and displayed news headlines.

TASK 5 -COVID-19 STATISTICS API INTEGRATION

CODE:

```
16
17 import requests
18 import time
19 def covid_stats(country):
20     url = f"https://disease.sh/v3/covid-19/countries/{country}"
21     for attempt in range(3):
22         try:
23             response = requests.get(url, timeout=5)
24             if response.status_code == 200:
25                 data = response.json()
26                 print("\n--- COVID-19 Statistics ---")
27                 print("Country:", data.get("country", "N/A"))
28                 print("Total Cases:", data.get("cases", "N/A"))
29                 print("Total Deaths:", data.get("deaths", "N/A"))
30                 print("Recovered:", data.get("recovered", "N/A"))
31                 print("Active Cases:", data.get("active", "N/A"))
32                 return
33             elif response.status_code == 404:
34                 print("Invalid country name.")
35                 return
36             elif response.status_code == 429:
37                 print("Rate limit exceeded. Waiting...")
38                 time.sleep(3)
39             else:
40                 print("API Error:", response.status_code)
41         except requests.exceptions.RequestException:
42             print("Network error... Retrying")
43             time.sleep(2)
44         print("Failed after multiple attempts.")
45     covid_stats("India")
```

1. sunt aut facere repellat provident occaecati excepturi optio reprehenderit
2. qui est esse
3. ea molestias quasi exercitationem repellat qui ipsa sit aut
4. eum et est occaecati
5. nesciunt quas odio

--- COVID-19 Statistics ---
Country: India
Total Cases: 45035393
Total Deaths: 533570
Recovered: 0
Active Cases: 44501823
PS C:\Users\Sameera Khan\OneDrive\Desktop\AI - LAB>

OBSERVATION:

The program successfully retrieved real-time COVID-19 statistics using an API. It handled errors such as invalid country names, network failures, and rate limits using retry logic. The data was displayed in a clear and structured format.